



Project 1 - Data Warehouse Design

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NOTE: For this project, you can work individually or in a team of a **maximum** of two people. CITS3401 students should form groups with other CITS3401 students. CITS5504 students should form groups with other CITS5504 students. Cross-unit grouping is only allowed on rare occasions, evaluated case by case. Project 1 contributes **25%** to the total assessment of this unit. The submission deadline is **17 April at 11:59 PM on LMS**.

The overall objectives of this project are to build a data warehouse using real-world datasets and to carry out a basic data mining activity, in this case, association rule mining.

Datasets and Problem Domain

According to IATA's analysis, the worldwide aviation industry experienced moderate growth in 2025. Passenger demand, measured by Revenue Passenger Kilometres, grew 5.3% compared to the previous year, slightly outpacing the 5.2% increase in airline capacity (Available Seat Kilometres). This imbalance drove the Passenger Load Factor to an unprecedented 83.6%, marking the highest annual utilisation rate ever recorded.

The growth trajectory showed divergence between market segments. International routes demonstrated robust expansion at 7.1% year-on-year, whilst domestic markets grew at a more modest 2.4% rate.

Regional Performance Highlights

Asia Pacific Region Carriers in the Asia Pacific region achieved the second-highest growth rate globally, with passenger traffic expanding 7.8% year-on-year.

Market-Specific Trends

Australia: The Australian domestic aviation market showed growth of 2.9% in passenger traffic (note: an earlier figure in the report cited 2.2% growth). However, airlines aggressively expanded capacity, resulting in a utilisation decline of 2.8 percentage points, bringing the load factor down to 82.2%.

The contrasting approaches between China and Australia highlight different strategic responses to post-pandemic recovery, with China prioritising load factors whilst Australia emphasised capacity restoration.

Assuming you are a business analyst, use the given data sources within a data warehouse to deliver insights to stakeholders, including governments, airline companies, or passengers.

Data Sources

12MB

Airline Dataset Updated - v2.csv

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Airline Dataset

The dataset is from [Kaggle](#).

Data dictionary

- **Passenger ID** - Unique identifier for each passenger
- **First Name** - First name of the passenger
- **Last Name** - Last name of the passenger
- **Gender** - Gender of the passenger
- **Age** - Age of the passenger

- **Nationality** - Nationality of the passenger
- **Airport Name** - Name of the airport where the passenger boarded
- **Airport Country Code** - Country code of the airport's location
- **Country Name** - Name of the country the airport is located in
- **Airport Continent** - Continent where the airport is situated
- **Continents** - Continents involved in the flight route
- **Departure Date** - Date when the flight departed
- **Arrival Airport** - Destination airport of the flight ([Kaggle ↗](#)'s explanation is incorrect, `Arrival Airport` is not a destination — it's the IATA code of the same airport as `Airport Name`).
 - This is identified as an **ETL transformation error**, where the pipeline appears to have populated `Arrival Airport` by converting `Airport Name` to its IATA code, rather than sourcing it from a genuine destination field.
 - **Strategy 1 — Rename the Column.** If the IATA code of the departure airport is considered analytically useful, the column should be retained but renamed to `Departure Airport Code` to accurately reflect its content. This preserves the data while correcting the misleading label.
 - **Strategy 2 — Delete the Column.** If the departure airport's IATA code adds no analytical value beyond what `Airport Name` already provides, the column should be dropped entirely during the transformation stage to reduce redundancy and avoid confusion in downstream analysis.
- **Pilot Name** - Name of the pilot operating the flight
- **Flight Status** - Current status of the flight (e.g., on-time, delayed, canceled)



airports.csv

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Airports

 24KB	countries.csv	 Download	 Open
Countries			
 1MB	navaids.csv	 Download	 Open
Navaids			
 468KB	regions.csv	 Download	 Open
Regions			

These four datasets are sourced from [OurAirports](#). Visit the website for more details.

Special Requirements for CITS5504 and CITS3401

- **For CITS3401 students**, you can use the Airline Dataset only. You are encouraged to use one or more of the additional provided datasets.
- **CITS5504 students** are required to use the Airline Dataset along with at least one additional provided dataset. You must also build a data warehouse using both PostgreSQL and Databricks. The schema for the Databricks data warehouse can be straightforward — a fact table with a few dimension tables.
- **All students** are encouraged to use external datasets associated with the Airline Dataset.

Data Warehousing Design and Implementation

Design a data warehouse for the dataset(s) by following the four steps below of dimensional modelling (i.e., Kimball's four steps).

1. Identify the process being modelled.
2. Determine the grain at which facts can be stored.
3. Choose the dimensions
4. Identify the numeric measures for the facts.

To realise the four steps, we can start by drawing and refining a StarNet with the above four questions in mind.

1. Think about a few business questions that your data warehouse could help answer.
2. Draw a StarNet to identify the dimensions and concept hierarchies for each dimension. This should be based on the lowest level of information you have access to.
3. Use the StarNet footprints to illustrate how the business queries can be answered with your design. Refine the StarNet if the desired queries cannot be answered, for example, by adding more dimensions or concept hierarchies.
4. Once the StarNet diagram is completed, draw it using software such as Microsoft Visio (free to download under [Azure Education ↗](#)) or a drawing program/[website ↗](#) of your own choice. A Hand-printed StarNet diagram is also welcome!
5. Implement a star or snowflake schema using PostgreSQL. For the fact table and dimension tables, clearly state which ones are measures and dimensions, and indicate the dimension references.
6. Use PostgreSQL to build a multi-dimensional analysis service solution, with a cube designed to answer your business queries. Make sure the concept hierarchies match your StarNet design.

7. Use Power BI/Tableau to visualise the data returned from your business queries.

Association Rule Mining

 **NOTE:** you are not allowed to use Microsoft Visual Studio, R, or Weka to do association rule mining in this project. Only Python code will be considered (for association rule mining) in this project.

Make sure you complete the relevant lab before attempting this task. The lab content may help you complete this task.

You are expected to select a subset (or all) of the attributes to mine interesting patterns. To rank the interestingness of the extracted rules, use support, confidence, and lift.

In the submitted PDF, you need to:

- Explain and discuss which association rules mining algorithms were used with references.
- Explain the top k rules (where $k \geq 1$) that have "**Flight Status**" on the right-hand side only, ranked by lift and confidence.
- Explain the meaning of these k rules in plain English.
- Share insights derived from the mining results. If no meaningful rules are discovered, explore potential reasons for this outcome.
- Based on the rules, provide and explain at least THREE (3) recommendations to the government on civil aviation industry development.

What to Submit

The following are the files needed for Project 1 submission.

⚠ NOTE:

1. In-text citations and end-text references are required in this project report, following the [IEEE referencing style](#) ⁷.
2. Include team members' information in the report, such as team members' names and student IDs.

1. A comprehensive PDF report that includes the following key components:

- **Design, implementation, and usage of the data warehouse:** Explain and discuss the design, implementation, and usage of the data warehouse to answer your queries, such as the fact table, dimension tables, and concept hierarchies design. When designing your data warehouse, you may select any number of columns/attributes as dimensions and measures. Please justify your design by explaining your assumptions or reasoning behind your choices with references.

NOTE: The data warehouse design should demonstrate concept hierarchies and good dimension modelling. An overly simple design will result in low marks for the project.

- **Schema, Starnet and query footprints:** Provide and explain your StarNet diagrams and the query footprints. Discuss which schema you used and justify your choice with references.
- **Data cleaning, preprocessing, and ETL process:** Describe your ETL process in detail. Include descriptions of the techniques used, discuss your ETL principles, explain the reasoning behind the key steps, and provide screenshots illustrating the process flow with references.

NOTE: During the ETL process, you may remove some rows from the main dataset . However, the number of dropped rows should not exceed 5% of the Airline dataset (Airline Dataset Updated - v2). Removing too many rows will result in a low mark for the ETL section.

- **Visualisation of query results:** Present the findings from your business queries using appropriate charts, graphs, and other visualisations. Ensure the insights are clearly communicated and easily understandable to stakeholders.
- **Association rules mining:** See the [Association Rules Mining section](#) above.

2. All scripts you used (e.g. Python, PostgreSQL) with clear and well-structured comments.
3. The Power BI/Tableau file (.pdf file).
4. All CSV files you used for building and populating your database.

 **NOTE:** All files need to be zipped up in a **single zip file** and submitted to [LMS](#)  (Under the **Projects channel** that can be found in the left navigation panel).

Marking Scheme

Scheme	Marks
Schema of each dimension and concept hierarchies for each dimension	5
Corresponding StarNet to illustrate query capabilities of the data warehouse	5
At least 5 types of business queries that the StarNet can answer	5
Data warehouse schema and fact table design	5
Data cleaning/pre-processing/ETL process for data transformation (include code, screenshots, and detailed explanations)	10
SQL Script file for building the database and loading the datasets	5
Visualise corresponding to the 5 business queries with appropriate charts	5
Association rule mining meaningful set-up	5
Interpretation of top rules and suggestions	5
Coherence between the design and implementation, quality and complexity of the solution, reproducibility of the solution	5
Overall report quality	5
Total	60

Data warehousing exercises are often open-ended. In other words, there is almost always a better solution. You can interpret the scale of marks as:

5 - Exemplary (comprehensive solution demonstrating professional application of the knowledge taught in the class with initiative beyond just meeting the project requirement)

4 - Proficient (correct application of the taught concepts with clear understandings demonstrated)

3 - Satisfactory (managed to meet most of the project requirements)

2 - Developing (some skills are demonstrated but need revision)

1 - Not yet Satisfactory (minimal effort)

0 - Not attempted.

Can I Use Generation AI?

Informed and educated use of generative AI tools is encouraged and should be declared. If you choose to use generative AI, please document its suggestions in your report and provide a clear rationale for accepting or rejecting each recommendation. While using generative AI is optional, as a data science professional, you are strongly encouraged to develop critical evaluation skills and build familiarity with modern tools that could enhance your productivity.

 **NOTE:** Using generative AI without proper citation and documentation is unacceptable.

The following example is a BAD example that may risk receiving a 0 mark.

Association Rule Mining

The top k rules have suitable columns on the right-hand side based on a suitable metric, where $k \geq 1$, which means statistically significant or interesting on the right-hand side. Association rule mining assists with the identification of the most relevant or meaningful connection between them, based on support, confidence and lift metrics. When the value of $k \geq 1$, the correlation is high and it indicates a more meaningful connection together. In the case of the Olympic Games' relationship with mental health issues, there is a lack of meaningful values that can help with the analysis for the clients. It is because of the large number of cases that represent the dataset and the imbalance comparison to the $k \geq 1$ comparison values.

Hallucinated wrong answer generated from LLM

In this case, $k > 1$ means the student should explain more than 1 top association mining rule. However, the student misunderstood the k value and the concept of lift. This is a common error when using GenAI, which sometimes generates non-existent theories - a phenomenon called '[AI hallucination](#)'.

The explanations below are correct understandings.

Y Explain K rules in Association rules mining

K Rules in Association Rule Mining

In association rule mining, "k rules" refers to the practice of finding and analyzing the top k association rules from a dataset based on specific quality metrics. Here's an explanation of this concept:

What are k rules?

When mining association rules, algorithms often generate a large number of potential rules. The "k rules" approach means selecting the top k rules according to a chosen evaluation metric, such as:

Correct Explanation

Explain the top k rules (where $k \geq 1$) that have "Road User" on the right-hand side, ranked by lift and confidence. What does K mean?

What Does k Mean?

In association rule mining, k represents the number of top rules to extract. When we say **top k rules**, we mean the k most interesting association rules based on certain ranking metrics.

Understanding the Question:

- We are looking for rules where "Road User" appears on the **right-hand side (RHS)**.
- **Ranking criteria:** The rules are ranked by **lift** (which measures strength) and **confidence** (which measures reliability).
- $k \geq 1$: The number of top rules to extract must be at least 1.

Example of Association Rules

Consider a dataset of traffic incidents. Some example rules could be:

Rule	Confidence	Lift
{Vehicle Type = Bicycle} $\rightarrow$ {Road User}	90%	2.5
{No Helmet} $\rightarrow$ {Road User}	85%	2.3
{Traffic Violation = Speeding} $\rightarrow$ {Road User}	78%	2.1

- If $k = 2$, we extract the top 2 rules:
 1. {Vehicle Type = Bicycle} $\rightarrow$ {Road User} (highest lift and confidence)
 2. {No Helmet} $\rightarrow$ {Road User}

Correct Explanation

Last updated 15 days ago